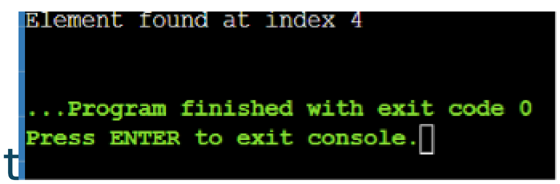
Binary Search — Definition

Binary Search is a searching algorithm used to find an element in a **sorted array/list**. It repeatedly divides the search range into half and eliminates the half that cannot contain the target.

Algorithm

1. Start with two pointers:
   a. `low = 0`
   b. `high = n - 1`
2. Find the middle element:
   `mid = (low + high) // 2`
3. Compare `arr[mid]` with the target:
   a. If equal → **element found**.
   b. If target is greater → search the **right half**.
   c. If target is smaller → search the **left half**.
4. Repeat until the element is found or `low > high`.
5. If `low > high`, the element is **not present**.

output



Time Complexity

Case	Complexity
Best Case	**O(1)**
Average Case	**O(log n)**
Worst Case	**O(log n)**

Space Complexity

- **Iterative Binary Search: O(1)**
- **Recursive Binary Search: O(log n)** because of the recursion stack.

Important: Binary search requires the array/list to be **sorted**.